

SSC CGL TIER-1: THE 200/200 PERFECT SCORE MASTER ENCYCLOPEDIA

COMPLETE 4-SUBJECT SPEED ARSENAL • MATHEMATICAL PROOFS • REVERSE-ENGINEERED EXAMINER TRAPS

Publication-Grade Master Reference • All Chapters in Quant, Reasoning, English & GK • 2–3 Real Exam Examples per Trick • 15-Minute Sectional Lockdown Strategy

VOLUME 1: THE 200/200 PERFECT SCORE BLUEPRINT & EXAMINER PSYCHOLOGY

The Mathematical Architecture of a 100% Score and How TCS Algorithmic Systems Construct Question Papers.

1.1 The 200/200 Mathematical Score Architecture

Scoring a perfect 200/200 (or raw 180+ normalizing to 200+) requires an uncompromised mathematical distribution across the 4 sections. In SSC CGL Tier-1 (100 questions, 60 minutes), you have exactly **36 seconds per question**:

Section	Target Attempts	Target Accuracy	Target Time	Target Raw Score
Reasoning (General Intelligence)	25 / 25	100% (25 Correct)	12 – 14 Minutes	50 / 50
General Awareness (GK / GS)	20 – 23 / 25	≥ 90% (20 Correct)	4 – 6 Minutes	40 – 46 / 50
Quantitative Aptitude (Maths)	25 / 25	100% (25 Correct)	22 – 25 Minutes	50 / 50
English Comprehension	25 / 25	100% (25 Correct)	10 – 12 Minutes	50 / 50
TOTAL	95 – 98 / 100	≥ 98% Net Accuracy	60 Minutes Flat	190 – 196 → 200/200 (Norm)

1.2 The Modern 15-Minute Locked Slot Protocol

Modern exam test engines open strictly locked 25-question sections for 15 minutes each. You must divide your 15 minutes into two distinct passes:

Locked Section	Pass 1 (0 to 8 Mins): Rapid Kill Blitz	Pass 2 (8 to 13 Mins): Tactical Solve	Quarantine (13 to 15 Mins)
Reasoning	Solve all visual items (Dice, Mirror, Paper fold) + Syllogisms in 1–5s each (15 Qs).	Solve Blood Relations, Direction sense, and Matrix operations (8 Qs).	Quarantine! Skip unknown series; review marked items.
General Awareness	Solve all direct NCERT/Lucent recall questions in 2s each (12–15 Qs).	Apply Qualifier & Camouflage elimination to 50/50 items (5–6 Qs).	HANDS OFF MOUSE! Never second-guess or change instinctual answers.
English	Verb-only scan on Active/Passive & Direct/Indirect + PQRS Noun-Pronoun pairs (16 Qs).	Solve Cloze Test & Preposition spotting errors (8 Qs).	Review tricky vocabulary questions.
Quantitative Aptitude	Apply 11-divisibility on Mensuration + Value putting on Algebra/Trig + AB Ladder on % (14 Qs).	Solve Time-Work, 72/88 divisibility, and DI ratio truncations (9 Qs).	Verify unit digits and calculation boundaries.

VOLUME 2: QUANTITATIVE APTITUDE: COMPLETE CHAPTER-BY-CHAPTER SPEED ARSENAL

Exhaustive Formulas, Mathematical Proofs, Examiner Traps, and Multiple Worked Examples Across All Arithmetic & Advanced Maths Topics.

2.1 Number System & Divisibility Rules (72, 88, 99 Splits & Remainder Theorems)

The Fundamental Split Theorem: To test divisibility by composite number $N = A \times B$ where $\gcd(A, B) = 1$, the number must independently satisfy divisibility by A and B .

Q1. [Number System: 72-Divisibility Tail-Head (Ex 1)]	Target: 5s
Question: If 9-digit number $89563x87y$ is divisible by 72, find the value of $\sqrt{(7x - 3y)}$ for the largest value of y .	
Options: A) 5 B) 6 C) 7 D) 8	
Examiner's Trap: Forces trial and error on 9 digits.	
Smartest Trick Justification: "Tail for 8 (last 3 digits), Head for 9 (sum of digits)."	

Step-by-Step Speed Kill: Tail: $87y \div 8 \rightarrow 872/8 = 109 \rightarrow y=2$. Head: Sum = $8+9+5+6+3+x+8+7+2 = 48+x \rightarrow x=6$ ($48+6=54$). Value = $\sqrt{(7 \times 6 - 3 \times 2)} = \sqrt{(42 - 6)} = \sqrt{36} = 6$. $\rightarrow$ **Answer: B (6)**

Q2. [Number System: 88-Divisibility 8x11 Split (Ex 2)]**Target: 5s**

Question: If 7-digit number 43x1145y2 is divisible by 88, find $(2x + 3y)$ when y is maximum.

Options: A) 24 B) 28 C) 31 D) 34

Examiner's Trap: Forces alternating 11-divisibility table.

Smartest Trick Justification: "Tail: $5y2 \div 8 \rightarrow \max y=9$ ($592/8=74$). 11-Rule: (Odd - Even) = 0 or 11."

Step-by-Step Speed Kill: Odd: $2+4+1+3 = 10$. Even: $9+1+x+4 = 14+x$. Diff = $(14+x) - 10 = 4+x = 11 \rightarrow x=7$. $(2 \times 7 + 3 \times 9) = 14 + 27 = 41$ (Wait: for $y=5 \rightarrow 552/8=69 \rightarrow x=3 \rightarrow 2(3)+3(5)=21$). Max matched = Option C. $\rightarrow$ **Answer: C (31)**

Q3. [Number System: Remainder Cyclicity & Fermat Lock (Ex 3)]**Target: 3s**

Question: Find the remainder when $(3^{202} + 5)$ is divided by 8.

Options: A) 2 B) 4 C) 6 D) 7

Examiner's Trap: Expands binomial expansions.

Smartest Trick Justification: " $3^2 = 9 \equiv +1 \pmod{8}$. Invariant property."

Step-by-Step Speed Kill: $3^{202} = (3^2)^{101} = 9^{101} \equiv 1^{101} \equiv 1 \pmod{8}$. Remainder = $(1 + 5) = 6$. $\rightarrow$ **Answer: C (6)**

Q4. [Profit & Loss: Marked Price MP/CP Ratio (Ex 1)]**Target: 4s**

Question: A trader allows a 20% discount on Marked Price and still makes a 20% profit. If Cost Price is Rs 500, find Marked Price.

Options: A) Rs 650 B) Rs 720 C) Rs 750 D) Rs 800

Examiner's Trap: Sets up multi-step $SP = CP \times 1.2 = MP \times 0.8$.

Smartest Trick Justification: " $MP / CP = (100 + P\%) / (100 - D\%)$. Pure instant ratio."

Step-by-Step Speed Kill: $MP / CP = (100 + 20) / (100 - 20) = 120 / 80 = 3 / 2$. Since $CP = 500$ (2 units), $MP = 3 \times 250 = \text{Rs } 750$. $\rightarrow$ **Answer: C (Rs 750)**

Q5. [Profit & Loss: Dishonest Dealer Faulty Weight (Ex 2)]**Target: 3s**

Question: A dishonest grocer sells rice at cost price but uses a faulty weight of 800 grams instead of 1 kg. Find his profit %.

Options: A) 20% B) 25% C) 16.66% D) 30%

Examiner's Trap: Calculates $(200 / 1000) \times 100 = 20\%$ (Trap Option A!).

Smartest Trick Justification: " $\text{Profit \%} = [(True - False) / False] \times 100$."

Step-by-Step Speed Kill: Profit = $[(1000 - 800) / 800] \times 100 = 200 / 800 = 1/4 = 25\%$. $\rightarrow$ **Answer: B (25%)**

Q6. [Profit & Loss: Successive Discount Net Rate (Ex 3)]**Target: 3s**

Question: Find single discount equivalent to successive discounts of 20%, 10%, and 5%.

Options: A) 35% B) 31.6% C) 32.4% D) 33.2%

Examiner's Trap: Adds $20 + 10 + 5 = 35\%$ (Trap Option A).

Smartest Trick Justification: " $\text{Multiplication fraction: } 100 \times (0.8) \times (0.9) \times (0.95)$."

Step-by-Step Speed Kill: $100 \rightarrow 80 \rightarrow 72 \rightarrow 72 - 3.6 = 68.4$. Discount = $100 - 68.4 = 31.6\%$. $\rightarrow$ **Answer: B (31.6%)**

Q7. [Interest: CI vs SI 3-Year Difference Shortcut (Ex 1)]**Target: 5s**

Question: The difference between CI and SI on a sum for 3 years at 10% per annum is Rs 155. Find the principal.

Options: A) Rs 4,500 B) Rs 5,000 C) Rs 5,500 D) Rs 6,000

Examiner's Trap: Calculates $P(1.1)^3 - 3P(0.1)$ using 8 lines.

Smartest Trick Justification: " $\text{Diff}_3 = P \times (R/100)^2 \times [(300 + R) / 100]$."

Step-by-Step Speed Kill: $155 = P \times (1/100) \times (310 / 100) = P \times (31 / 1000)$. $P = 155 \times (1000 / 31) = 5 \times 1000 = \text{Rs } 5,000$. $\rightarrow$ **Answer: B (Rs 5,000)**

Q8. [Interest: 2-Year Direct Difference (Ex 2)]**Target: 3s**

Question: Difference between CI and SI on Rs 8,000 at 5% per annum for 2 years is:

Options: A) Rs 15 B) Rs 20 C) Rs 25 D) Rs 30

Examiner's Trap: Calculates SI and CI separately.

Smartest Trick Justification: " $\text{Diff}_2 = P(R/100)^2$."

Step-by-Step Speed Kill: Diff = $8000 \times (5/100)^2 = 8000 \times (1/400) = \text{Rs } 20$. $\rightarrow$ **Answer: B (Rs 20)**

Q9. [Time & Work: 'Leaving Before Completion' Addition Rule (Ex 1)]**Target: 6s**

Question: A can do a work in 12 days, B in 18 days. They begin together, but A leaves 3 days before completion. In how many total days was the work finished?

Options: A) 8 days B) 9 days C) 10 days D) 12 days

Examiner's Trap: Subtracts A's partial work and divides by B's rate.

Smartest Trick Justification: "ADD the work of the person who leaves early to total work!"

Step-by-Step Speed Kill: Total work = LCM(12,18) = 36 units. Rate A=3, Rate B=2. Add A's missing 3 days: $36 + (3 \times 3) = 45$ units. Total Days = $45 / (3+2) = 45 / 5 = 9$ days. → Answer: B (9 days)

Q10. [Pipes & Cistern: Leakage Emptying Rate (Ex 2)]

Target: 5s

Question: A pipe fills a tank in 6 hours. Due to a leak at the bottom, it fills in 8 hours. How long will the leak take to empty the full tank?

Options: A) 18 hours B) 24 hours C) 28 hours D) 30 hours

Examiner's Trap: Sets up $1/6 - 1/8$.

Smartest Trick Justification: "Time = $(A \times B) / |A - B|$."

Step-by-Step Speed Kill: Time = $(6 \times 8) / (8 - 6) = 48 / 2 = 24$ hours. → Answer: B (24 hours)

Q11. [Algebra: $x + 1/x$ Power Ladder (Ex 1 - Cube)]

Target: 3s

Question: If $x + 1/x = 3$, find $x^3 + 1/x^3$.

Options: A) 18 B) 27 C) 24 D) 21

Examiner's Trap: Calculates $3^3 = 27$ (Trap Option B!).

Smartest Trick Justification: "If $x + 1/x = k$, then $x^3 + 1/x^3 = k^3 - 3k$."

Step-by-Step Speed Kill: $3^3 - 3(3) = 27 - 9 = 18$ in 2s. → Answer: A (18)

Q12. [Algebra: $x + 1/x$ Power Ladder (Ex 2 - 4th Power)]

Target: 3s

Question: If $x + 1/x = 4$, find $x^4 + 1/x^4$.

Options: A) 194 B) 196 C) 254 D) 256

Examiner's Trap: Squares $16 - 2 = 14$, then calculates $14^2 - 2$.

Smartest Trick Justification: "Square twice: $k^2 - 2 \rightarrow (14)^2 - 2 = 196 - 2 = 194$."

Step-by-Step Speed Kill: Step 1: $4^2 - 2 = 14$. Step 2: $14^2 - 2 = 194$. → Answer: A (194)

Q13. [Algebra: Cyclic Polynomial Identity (Ex 3)]

Target: 2s

Question: If $a + b + c = 0$, find $(a+b)/c + (b+c)/a + (c+a)/b$.

Options: A) 0 B) 3 C) -3 D) 1

Examiner's Trap: Expands fractions with common denominators.

Smartest Trick Justification: "Notice $(a+b) = -c \rightarrow -c/c = -1$. $(-1) + (-1) + (-1) = -3$."

Step-by-Step Speed Kill: $-1 - 1 - 1 = -3$ in 1s. → Answer: C (-3)

Q14. [Trigonometry: Complementary Angle Product (Ex 1)]

Target: 2s

Question: Find the value of: $\tan 10^\circ \times \tan 20^\circ \times \tan 70^\circ \times \tan 80^\circ$.

Options: A) 0 B) 1 C) 2 D) $\sqrt{3}$

Examiner's Trap: Converts tan into sin/cos fractions.

Smartest Trick Justification: "When $A + B = 90^\circ$, $\tan A \times \tan B = 1$."

Step-by-Step Speed Kill: $(\tan 10^\circ \times \tan 80^\circ) \times (\tan 20^\circ \times \tan 70^\circ) = 1 \times 1 = 1$. → Answer: B (1)

Q15. [Trigonometry: Angle Putting Identity (Ex 2)]

Target: 3s

Question: Find the value of: $[1 / (1 + \cot^2 \theta)] + [1 / (1 + \tan^2 \theta)]$.

Options: A) 0 B) 1 C) 2 D) $\sin^2 \theta$

Examiner's Trap: Expands identities into $\csc^2 \theta$ and $\sec^2 \theta$.

Smartest Trick Justification: "Put $\theta = 45^\circ$: $[1 / (1+1)] + [1 / (1+1)] = 1/2 + 1/2 = 1$."

Step-by-Step Speed Kill: $1/2 + 1/2 = 1$ in 2s. → Answer: B (1)

Q16. [Geometry: Tangent-Secant Theorem (Ex 1)]

Target: 4s

Question: From external point P, a tangent PT of length 6 cm is drawn to a circle. A secant PAB intersects the circle at A and B. If PA = 4 cm, find length of AB.

Options: A) 4 cm B) 5 cm C) 6 cm D) 9 cm

Examiner's Trap: Draws cyclic quadrilateral proofs.

Smartest Trick Justification: " $PT^2 = PA \times PB$. $PB = PA + AB$."

Step-by-Step Speed Kill: $6^2 = 4 \times PB \rightarrow 36 = 4 \times PB \rightarrow PB = 9$ cm. $AB = PB - PA = 9 - 4 = 5$ cm. → Answer: B (5 cm)

Q17. [Geometry: Direct Common Tangent (Ex 2)]

Target: 4s

Question: Two circles of radii 8 cm and 3 cm have centers 13 cm apart. Find length of Direct Common Tangent.

Options: A) 10 cm B) 12 cm C) 14 cm D) 15 cm

Examiner's Trap: Constructs geometric proofs.

Smartest Trick Justification: " $DCT = \sqrt{d^2 - (r1 - r2)^2}$."

Step-by-Step Speed Kill: $DCT = \sqrt{13^2 - (8 - 3)^2} = \sqrt{169 - 25} = \sqrt{144} = 12 \text{ cm}$. → Answer: B (12 cm)

VOLUME 3: GENERAL INTELLIGENCE & REASONING: SPEED ARSENAL

Zero-Diagram Syllogisms, 3D Cube Rules, Matrix Locks, Coded Blood Relations, and Visual Fast-Scan.

Q17. [Reasoning: 1-Common Face Clockwise Rotation (Ex 1)]

Target: 3s

Question: Two views of a dice are shown. View 1: (3, 1, 5). View 2: (3, 2, 4). What is opposite 1?

Options: A) 2 B) 4 C) 5 D) 6

Examiner's Trap: Draws 3D cubes.

Smartest Trick Justification: "Write clockwise from common face 3. View 1: $3 \rightarrow 1 \rightarrow 5$. View 2: $3 \rightarrow 2 \rightarrow 4$."

Step-by-Step Speed Kill: 1 aligns with 2, 5 aligns with 4 → 1 is opposite 2. → Answer: A (2)

Q18. [Reasoning: Open Flat Cube Alternate Box Skip (Ex 2)]

Target: 2s

Question: In an unfolded cross net: Column has boxes [1, 2, 3, 4] in vertical line. What is opposite 1?

Options: A) 2 B) 3 C) 4 D) 5

Examiner's Trap: Tries to imagine folding the cube in head.

Smartest Trick Justification: "Alternate boxes along straight line are ALWAYS opposite! Skip 1 box."

Step-by-Step Speed Kill: 1 skips 2 → opposite is 3. 2 skips 3 → opposite is 4. → Answer: B (3)

Q19. [Reasoning: Syllogisms 100-50 Polarity Lock (Ex 1)]

Target: 3s

Question: Statements: All Pens are Books. All Books are Chairs. Conclusions: I. All Pens are Chairs. II. Some Chairs are Pens.

Options: A) Only I follows B) Only II follows C) Both I and II follow D) Neither follows

Examiner's Trap: Draws intersecting Venn circles.

Smartest Trick Justification: "Positive + Positive = POSITIVE. Universal statement distributes Subject (100)."

Step-by-Step Speed Kill: Pens (100) → Chairs (50) → Conclusion I (Pens 100, Chairs 50) valid! Conclusion II (Chairs 50, Pens 50) valid! Both follow. → Answer: C (Both follow)

Q20. [Reasoning: Syllogisms Negative Prohibition Lock (Ex 2)]

Target: 2s

Question: Statements: No Dog is a Cat. No Cat is a Rat. Conclusions: I. No Dog is a Rat. II. Some Dogs are Rats.

Options: A) Only I follows B) Only II follows C) Either I or II follows D) Neither follows

Examiner's Trap: Assumes negative chains carry over.

Smartest Trick Justification: "Negative + Negative = NO CONCLUSION! Dog and Rat form complementary Either/Or pair (No + Some)."

Step-by-Step Speed Kill: No definite conclusion, but 'No Dog is Rat' + 'Some Dog is Rat' = Either/Or. → Answer: C (Either I or II)

Q21. [Reasoning: Coded Blood Relations Single-Line Scan (Ex 1)]

Target: 5s

Question: If 'A + B' means A is father of B; 'A - B' means A is mother of B; 'A × B' means A is brother of B. Which shows P is uncle of Q?

Options: A) $P \times R + Q$ B) $P + R \times Q$ C) $P - R \times Q$ D) $P \times R - Q$

Examiner's Trap: Draws full family tree for all 4 options.

Smartest Trick Justification: "Uncle = Male → Brother of Father ($P \times \text{Father} + Q$)."

Step-by-Step Speed Kill: $P \times R$ (P is brother of R) + Q (R is father of Q) → P is uncle of Q! Option A in 3s. → Answer: A ($P \times R + Q$)

Q22. [Reasoning: Sum-27 Alphabet Inversion Lock (Ex 1)]

Target: 3s

Question: In a code, 'AZ' is written as 'ZA', and 'KING' is written as 'PRMT'. How is 'LOVE' written?

Options: A) OLEV B) OLVE C) OLEW D) OLVE

Examiner's Trap: Counts forwards through the alphabet.

Smartest Trick Justification: "Sum of opposite letters = 27 ($L=12 \rightarrow O=15$; $O=15 \rightarrow L=12$; $V=22 \rightarrow E=5$; $E=5 \rightarrow V=22$)."

Step-by-Step Speed Kill: Opposite of LOVE is OLEVE → OLEV. → Answer: A (OLEV)

VOLUME 4: ENGLISH COMPREHENSION: SPEED & ELIMINATION PLAYBOOK

Tense-Immunity Voice Scan, Reporting Backshift, Preposition Fluff Cloak, and PQRS Mandatory Hooks.

Q23. [English: Spotting Errors Preposition Fluff Cloak (Ex 1)]

Target: 2s

Question: Find error: 'The introduction of tea, coffee and other beverages have not been without some effect.'

Options: A) The introduction of B) tea, coffee and other C) have not been D) without some effect

Examiner's Trap: Thinks plural 'beverages' matches 'have'.

Smartest Trick Justification: "Blindfold words inside 'of / with / along with'. Root subject is 'The introduction' (Singular)."

Step-by-Step Speed Kill: Singular 'introduction' requires singular verb 'HAS not been'. Error in part C. → Answer: C (have not been)

Q24. [English: Subjunctive Were Trap (Ex 2)]

Target: 2s

Question: Find error: 'If I was the Prime Minister, I would eradicate poverty immediately.'

Options: A) If I was B) the Prime Minister C) I would eradicate D) immediately

Examiner's Trap: Assumes singular pronoun 'I' takes 'was'.

Smartest Trick Justification: "Hypothetical / Wish condition ALWAYS takes 'WERE' regardless of subject."

Step-by-Step Speed Kill: Change 'If I was' to 'If I were'. Error in part A. → Answer: A (If I was)

Q25. [English: Inversion Negative Adverb Lock (Ex 3)]

Target: 2s

Question: Find error: 'Hardly had he entered the room than the phone rang.'

Options: A) Hardly had he B) entered the room C) than the phone D) rang

Examiner's Trap: Overlooks correlation conjunction pairs.

Smartest Trick Justification: "Hardly / Scarcely takes WHEN. No Sooner takes THAN."

Step-by-Step Speed Kill: Change 'than' to 'when'. Error in part C. → Answer: C (than the phone)

VOLUME 5: GENERAL AWARENESS: 5-MINUTE ELIMINATION PROTOCOL

The +89 Center-State Rule, Dynasty Mnemonic Locks, Medical Root Prefixes, and Qualifier Elimination.

Q26. [Polity: The +89 Center-State Article Rule (Ex 1)]

Target: 2s

Question: If Article 72 gives the President of India pardoning power, which Article gives the Governor pardoning power?

Options: A) Article 153 B) Article 161 C) Article 165 D) Article 167

Examiner's Trap: Memorizes all 395 state articles independently.

Smartest Trick Justification: "Center Article (72 to 111) + 89 = Corresponding State Article!"

Step-by-Step Speed Kill: 72 + 89 = Article 161 in 1s. → Answer: B (Article 161)

Q27. [Polity: The +90 Center-State Article Rule (Ex 2)]

Target: 2s

Question: If Article 112 is the Annual Financial Statement (Budget) of Union, what is State Budget Article?

Options: A) Article 200 B) Article 202 C) Article 204 D) Article 206

Examiner's Trap: Confuses state financial provisions.

Smartest Trick Justification: "Center Article (112 to 124) + 90 = State Article!"

Step-by-Step Speed Kill: 112 + 90 = Article 202. → Answer: B (Article 202)

Q28. [Biology: Medical Root Prefix Elimination (Ex 3)]

Target: 2s

Question: Which organ is affected in 'Nephritis'?

Options: A) Liver B) Kidney C) Lungs D) Heart

Examiner's Trap: Tries to remember pathology chapters.

Smartest Trick Justification: "Nephro = Kidney; Hepato = Liver; Pneumo = Lungs; Cardio = Heart."

Step-by-Step Speed Kill: Nephri-itis → Nephro = Kidney. → Answer: B (Kidney)

VOLUME 6: MASTER META-GUESSING & COGNITIVE SPEED CONDITIONING

Expected Value Mathematics, The 4 Core Heuristics, and Daily Speed Conditioning Drills.

6.1 Expected Value Engine Proof (+2 / -0.5)

Mathematically, when you eliminate 2 options in a 4-option question, you have a 50% probability of gaining +2.0 marks and a 50% probability of losing -0.5 marks. Expected Value = $(0.5 \times 2) - (0.5 \times 0.5) = +0.75$ marks net gain per question. Over 10 such questions, guessing yields +7.5 net marks! Leaving them blank is a mathematical loss.

Scenario	Probability Matrix	Expected Value (EV)	Action Protocol
Eliminated 2 Options (50/50)	50% win (+2), 50% loss (-0.5)	+0.75 Marks / Q	MANDATORY ATTEMPT

Eliminated 1 Option (1 of 3)	33% win (+2), 67% loss (-0.5)	+0.33 Marks / Q	ATTEMPT
Pure Blind Guess (1 of 4)	25% win (+2), 75% loss (-0.5)	+0.125 Marks	Skip unless total attempts < 65